

Math 2200 (Calculus I) Section 05 Syllabus—Fall 2019

Instructor: Professor Bryce Christopherson
Office: Ross Hall 315
Email: bchris19@uwyo.edu

Office Hours: MTWF 11:30 AM–12:30 PM
in the CASM (RH 29)
Class: MTWF 10:00 AM – 10:50 AM in EN 3070

Course Supervisor: Regarding any difficulties regarding either the content or the administration of this course, you should first consult your instructor. If the instructor is not available or unable to help, you should next contact the Math 2200 Course Supervisor:

Dr. Eric Moorhouse, Ross Hall 216, (307)-766-4394, calculus@uwyo.edu.

Prerequisites: The prerequisite for Math 2200 is either a C or better in MATH 1405 or MATH 1450, a Mathematics Placement Exam score of 5 within one year prior to the start of the course, or ACT math score of 27. You must earn a C or better in this course to be admitted in Calculus II.

Textbook and Software: For this course you will need:

- Access to Pearson/MyLab Math (MLM). By default, you will register for this access through the WyoCourses site for your section of Math 2200 and you will be billed automatically. In the exceptional event that you have alternative access to MyLab Math, you will have the opportunity to opt out of this direct billing agreement during the first two weeks of classes but after a couple of weeks; afterwards, opting out will no longer be possible.
- A scientific non-graphing calculator such as a TI-30X.
- A physical copy of the textbook, *Calculus, Early Transcendentals* by Briggs & Cochran, 3rd Edition, is *optional*, but generally recommended. You will have access to an electronic version through MyLab/Math whether you purchase the physical book or not. *You are required to read all relevant sections of the textbook preferably before coming to class.*

Exams: All four exams are common to all coordinated sections of Math 2200 and will be administered outside the regularly scheduled class time (see below). Attendance is required and a make-up exam will only be administered only if there is documentation from a proper authority, such as a note from a physician in the case of illness. If you know that you will be missing an exam, please talk to your instructor or the course coordinator at least one week before the exam will be administered. If you have any questions about the exam time, please discuss this with your instructor or the course coordinator (calculus@uwyo.edu).

On Exams, you may use a scientific non-graphing calculator. Graphing calculators and notes may not be used on exams under any circumstances. The final exam will be comprehensive. Midterm Exams I, II and III, and the Final Exam are scheduled at the dates/times listed below; and locations will be announced later during the semester.

Homework:

Written Homework: Almost every Wednesday (subject to the discretion of your instructor), you will be given a short, in-class quiz which may be taken (if you desire) jointly with one or two partners. If you opt to take a quiz jointly in this fashion, all members of the group will receive the same grade for the quiz. To receive full credit, you will be required to show *all* of your work in a *neat, coherent, and mathematically correct* fashion. Notes, textbooks, and other sources may not be consulted during a quiz. If you are unable to attend class on a day that a quiz will be given, you must notify your instructor in advance. In cases where proper notification has not been given, or if the reason for missing the quiz is not valid, a make-up quiz will not be given.

MyLab/Math Homework: Online homework will be administered through your account with Pearson MyLab/Math. It is your responsibility to login to the site regularly and mind the due dates for these assignments, typically one assignment for every 1-2 sections covered in the textbook (every couple of days). On these assignments you will be permitted an unlimited number of tries on each question, using any aids available. Please take advantage of this opportunity to honestly learn the material (i.e. don't randomly guess at answers, or accept answers directly from other students).

Optional Quizzes/Review in MyLab/Math: In addition to online homework, a set of optional review quizzes and practice tests are available through the Pearson MyLab/Math software. These are provided to help you in your personal review, and will not count towards your grade in the class. Again, please take them seriously as a means to preparing for the exams.

Grading Policy: Your percentage grade is determined by the following:

Exam 1	100 pts	Thurs, Sept. 26, 5:15 PM–7:00 PM
Exam 2	100 pts	Thurs, Oct. 24, 5:15 PM–7:00 PM
Exam 3	100 pts	Thurs, Dec. 5, 5:15 PM–7:00 PM
Final Exam	100 pts	Tues, Dec. 17, 3:30 PM–5:30 PM
MyLab/Math Homework	75 pts	
Written Homework	125 pts	
Total	600 pts	

You can estimate your letter grade by using the following scale: 90–100% is an A, 80–89% is a B, 70–79% is a C, 60–69% is a D, and 0–59% is an F.

Plus/Minus Grades will not be awarded. You must achieve a C or higher to use this course as a prerequisite for other math courses.

How to be Successful in Calculus:

Time Management: Calculus I is a four credit class. This means you should be spending 8 to 12 hours a week on this course outside of the time in class. Most weeks you will be spending this time on doing your MyLab/Math homework and studying for quizzes/exams.

MyLab/Math: Students are sometimes tempted to use online calculators and computational engines to compute their online homework for them. Those that use such resources are not using MyLab/Math in a way that will help them to learn. Getting a few problems wrong on MyLab/Math and then asking for help on that problem will help you to master the material to do well on the exams.

Written Homework: The primary purpose of written homework is to provide feedback on your mathematical writing skills. When you submit your work, make sure that your work is organized, properly spaced out and easy to read.

Getting Help: One of the most important skills a student can learn is recognizing early that they are confused with some concept or skill in the class. Once you recognize you need help, first check through your materials, such as class notes, to see if the answer is there. If you cannot solve it yourself, *please seek help*. There are lots of resources available to you to help you succeed.

Study Groups: Meet with some of your classmates and help each other with questions. Learning from your peers is one of the best ways to learn.

Office Hours: My office hours (subject to change as noted above) 11:30 AM–12:30 PM, MTWF in the CASM (RH 29). Please seek me outside of class if at all possible; and if these times are not convenient, we should be able to arrange an alternative time. I am here to help you.

The Center for Assistance in Statistics and Mathematics (CASM): The CASM is a free drop-in tutoring center for students enrolled in 1000 and 2000 level math courses. It is located in Ross Hall 29 (northwest corner, on bottom floor). They are open Monday–Thursday 10:00 AM–5:00 PM and Friday 10:00 AM–1:00 PM. It's also a great place to study, work on homework, or meet with study groups. They have copies of current textbooks for students to borrow while there as well as several desktop computers, a printer, and two study rooms.

Supplemental Instruction(SI): SI is a series of out-of-class study sessions led by a student who has successfully completed the calculus sequence. The SI leader for this course is Taylor Hansen (thanse26@uwyo.edu). Meeting times and location will be announced by email early during the semester.

STEP Tutoring: The STEP Tutoring Program (<http://www.uwyo.edu/step>) offers drop-in one-on-one tutoring (no appointments) for Calculus students in Coe Library. Hours are Sunday–Thursday from 5:00 PM–9:00 PM. Check in at the Research Help Desk on the main level. Online tutoring (**eTutoring**) is also offered. For more information visit <http://www.uwyo.edu/studentaff/step/tutoring/>

Walk-In Academic Coaching The Center for Advising and Career Services in 222 Knight Hall offers walk-in coaching to any student. They can help on a variety of topics, such as note taking, time management, exam preparation, textbook reading, staying motivated, managing finals, etc.

STEP Satellite in Washakie: The STEP Satellite, located in the lower level of the Washakie Center, offers free academic support services to students on a drop-in basis Sunday through Thursday. Please see <http://www.uwyo.edu/step/> for more information.

It is important that you try a variety of different services. If you do not receive the help you need at that time, please try it again or try a different service. Don't be shy. Make sure you get the help you need!

Course Supervisor: Come to me if you are unhappy about some aspect of the course. In the event that a problem remains unresolved after our discussions, talk to Eric Moorhouse, the Math 2200 Supervisor (Office: RH 216, Phone: (307)-766-4394, Email: calculus@uwyo.edu).

Goals of Math 2200: This course fulfills the Quantitative Reasoning 2 (QB) or Qualitative (Q) requirement of the University Studies Program. QB and Q courses develop a student's numerical, logical, geometric, algorithmic and critical thinking skills as well as their ability to integrate these ways of thinking with verbal, written and creative thinking skills. Students will demonstrate mathematical and logical skill needed to formulate, analyze and interpret quantitative arguments in a variety of settings.

Calculus, one of the classical topics in mathematics, is the study of change. It is useful both in scientific fields and in applied studies from engineering to the life sciences. The primary goals of this course are to master the fundamental concepts and techniques of differential calculus in one variable, and to develop problem solving and critical thinking skills. By the end of this course, students should be able to

- Use algebraic, graphical and numerical skills and thinking to solve problems that involve limits and derivatives.
- Apply differential calculus concepts to a variety of applications.
- Manipulate and compare graphical, numerical and algebraic representations of mathematical relationships involving limits and derivatives.
- Calculate integrals using both the definition of the integral, and the Fundamental Theorem of Calculus.
- Manipulate and compare graphical, numerical and algebraic representations of mathematical relationships.
- Read and understand mathematics, think critically, and express mathematical concepts precisely in writing.
- Apply the knowledge gained in this course to other situations and disciplines.
- Be prepared to take Calculus II.

Academic Dishonesty, Classroom Conduct, Disability Student Services, etc.: Please Refer to the links provided through the WyoCourses site.

Changes to Syllabus: All deadlines, requirements, policies, and course structure is subject to change if deemed necessary by the instructor. Students will be notified of minor changes verbally in class and substantive changes shall be communicated in writing distributed either by an announcement posted to our class WyoCourses page or via email.

Grading Rubric: The rubric below will be used *as a guideline* when evaluating your written work on homework assignments and exams throughout this course. You should strive to produce written solutions that are clear and well organized, make proper use of mathematical notation and terminology, demonstrate an understanding of the mathematical concepts involved, and result in a correct solution. If you practice these skills while working on your written homework problems, you will be more likely to meet the standard expected of your written work on exams. Please familiarize yourself with this rubric and reference it if ever you feel your work has been unfairly graded.

Characteristics of the Written Work	Qualitative Description	Typical Scoring %
(i) The work is presented in a neat, clear, organized fashion that is complete and easy to follow. (ii) Graphs are correct, complete, carefully drawn and appropriately labeled. (iii) Correct terminology and notation are always used. (iv) The solution shows a complete understanding of the mathematical concepts involved, although it may contain a minor arithmetic error or sign mistake.	Exceptional	90-100
(i) The work may not be completely clear or well organized, or it may have some steps missing, but it is still possible to follow with little effort. (ii) Graphs are correct and complete, but somewhat carelessly drawn or missing some labels. (iii) Correct terminology and notation are usually used. (iv) The solution shows almost complete understanding of the mathematical concepts and may contain one or two arithmetic or sign errors.	Good	75-90
(i) The work may be unclear, poorly organized, or have several steps missing, but it is possible to deduce the intended method of solution with some effort. (ii) Graphs are partially correct, somewhat carelessly drawn, and may be missing some labels. (iii) Correct terminology and notation are sometimes used. (iv) The solution shows a limited understanding of the mathematical concepts involved, or an almost complete understanding of the mathematical concepts involved, but contains several arithmetic or sign errors.	Fair	60-75
(i) The work is poorly organized or is very incomplete, making it difficult to follow. (ii) Graphs are incorrect, carelessly drawn, or incomplete, and may not be labeled. (iii) Correct terminology and notation are seldom used. (iv) The solution shows a limited understanding of the mathematical concepts involved, and may also contain several arithmetic or sign errors.	Poor	40-60
(i) The work is so disorganized or incomplete that it is very difficult or impossible to follow. (ii) Graphs are not drawn or are completely wrong and not labeled. (iii) There is little or no correct usage of terminology and notation. (iv) The solution shows little or no understanding of the mathematical concepts involved.	Very Poor	0-40

Course Schedule:

Sections Covered

2.1-2.6, 3.1-3.11, 4.1-4.9, 5.1-5.4, 6.1

Su.	Mo.	Tu.	We.	Th	Fr.	Sa.
1 Sept.	2 Sept.	3 Sept.	4 Sept.	5 Sept.	6 Sept.	7 Sept.
	Labor Day		Intro		Review	
8 Sept.	9 Sept.	10 Sept.	11 Sept.	12 Sept.	13 Sept.	14 Sept.
	2.1	2.2	2.3 <u>Quiz 1(2.1,2.2)</u>		2.3	
15 Sept.	16 Sept.	17 Sept.	18 Sept.	19 Sept.	20 Sept.	21 Sept.
	2.4	2.5 <u>MLM 2.1,2.2</u>	2.6 <u>Quiz 2 (2.3,2.4)</u>		2.6 <u>MLM 2.3</u>	
22 Sept.	23 Sept.	24 Sept.	25 Sept.	26 Sept.	27 Sept.	28 Sept.
	2.6 <u>MLM 2.4</u>	3.1 <u>MLM 2.5</u>	Review <u>MLM 2.6</u> <u>Quiz 3 (2.5,2.6)</u>	EXAM 1	3.1	
29 Sept.	30 Sept.	1 Oct.	2 Oct.	3 Oct.	4 Oct.	5 Oct.
	3.2	3.3	3.3 <u>MLM 3.1</u> <u>Quiz 4 (3.1,3.2)</u>		3.4	<u>MLM 3.2</u>
6 Oct.	7 Oct.	8 Oct.	9 Oct.	10 Oct.	11 Oct.	12 Oct.
	3.4, 3.5	3.5 <u>MLM 3.3</u>	3.6 <u>Quiz 5 (3.3,3.4)</u>		3.6 <u>MLM 3.4</u>	
13 Oct.	14 Oct.	15 Oct.	16 Oct.	17 Oct.	18 Oct.	19 Oct.
	3.7 <u>MLM 3.5</u>	3.7	3.8 <u>MLM 3.6</u> <u>Quiz 6 (3.5, 3.6, 3.7)</u>		3.9	<u>MLM 3.7</u>
20 Oct.	21 Oct.	22 Oct.	23 Oct.	24 Oct.	25 Oct.	26 Oct.
	3.9	3.10 <u>MLM 3.8</u>	Review <u>Quiz 7 (3.8, 3.9, 3.10)</u>	EXAM 2	3.11	<u>MLM 3.9</u>

Su.	Mo.	Tu.	We.	Th	Fr.	Sa.
27 Oct.	28 Oct.	29 Oct.	30 Oct.	31 Oct.	1 Nov.	2 Nov.
	3.11	4.1 <u>MLM 3.10</u>	4.2		4.2 <u>MLM 3.11</u>	
3 Nov.	4 Nov.	5 Nov.	6 Nov.	7 Nov.	8 Nov.	9 Nov.
	4.3	4.4 <u>MLM 4.1</u>	4.4 <u>QUIZ 8 (3.11, 4.1, 4.2)</u>	<u>MLM 4.2</u>	4.5	<u>MLM 4.3</u>
10 Nov.	11 Nov.	12 Nov.	13 Nov.	14 Nov.	15 Nov.	16 Nov.
	4.6	4.6 <u>MLM 4.4</u>	4.7 <u>Quiz 9 (4.3, 4.4, 4.5)</u>		4.7 <u>MLM 4.5</u>	
17 Nov.	18 Nov.	19 Nov.	20 Nov.	21 Nov.	22 Nov.	23 Nov.
	4.9 <u>MLM 4.6</u>	5.1	5.1 <u>Quiz 10 (4.6, 4.7, 4.8, 4.9)</u>	<u>MLM 4.7</u>	5.2	<u>MLM 4.9</u>
24 Nov.	25 Nov.	26 Nov.	27 Nov.	28 Nov.	29 Nov.	30 Nov.
	Snow Day	Snow Day	Thanksgiving Break	Thanksgiving Break	Thanksgiving Break	
1 Dec.	2 Dec.	3 Dec.	4 Dec.	5 Dec.	6 Dec.	7 Dec.
<u>MLM 5.1</u>	5.2	5.3	Review <u>Quiz 11 (5.1, 5.2, 5.3)</u> <u>MLM 5.2</u>	Exam 3	5.3/5.4	
8 Dec.	9 Dec.	10 Dec.	11 Dec.	12 Dec.	13 Dec.	14 Dec.
	5.4/6.1 <u>MLM 5.3</u>	6.1	Review <u>Quiz 12 (5.4, 6.1)</u> <u>MLM 5.4</u>		Review (Last Day) <u>MLM 6.1</u>	
15 Dec.	16 Dec.	17 Dec.	18 Dec.	19 Dec.	20 Dec.	21 Dec.
	Finals Week	Final Exam	Finals Week	Finals Week	Finals Week	